

Momentum Profits and Idiosyncratic Volatility: The Korean Evidence¹

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Structured Abstract

Purpose – This study focuses on the profitability of momentum trading in the Korean stock market. More specifically, an examination of the relationship between momentum returns and idiosyncratic volatility (IVol) is conducted to determine whether momentum profits can be explained by IVol.

Design/methodology/approach – We form portfolios based on their past performance and examine the momentum, or contrarian returns, as the difference between winning and losing portfolios. To confirm that the momentum strategy provides excess returns, we study the relationship between momentum returns and IVol. We also examine the Fama and French (1993) three-factor model to see whether systematic risk affects momentum profits. We control firm size, stock price, and turnover to determine robustness. Finally, we investigate a time-series relationship between aggregate IVol and momentum profits.

Findings – We illustrate that excess returns are obtained from a momentum strategy, not a contrarian strategy, in the Korean stock market. Momentum returns are higher among high IVol stocks, especially high IVol winners. Examining the Fama and French (1993) three-factor model, we find that momentum returns cannot be explained by systematic risk. The findings are robust after controlling for factors such as firm size, book-to-market ratio, and turnover. We confirm the effect of IVol on momentum returns by illustrating that a time-series relationship between momentum returns and aggregate IVol is positive.

Originality/value – This paper is among the first, to our knowledge, to examine the relationship between momentum profits and IVol in the Korean stock market, one of the mature financial markets. The findings in this study can be applied to better understand the sources of gains from the momentum strategy in international stock markets.

Keywords: Idiosyncratic volatility, Momentum profits, Investment strategy

JEL Classification: G12, G14

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1. Introduction

Trading strategies based on historical returns, such as contrarian or momentum trading, have consistently received attention from both finance practitioners and academics. DeBondt and Thaler (1985, 1988) observe stocks becoming poor performers after a three to five year term of good performance. They suggest that a contrarian strategy allows investors to earn excess returns. On the other hand, Jegadeesh and Titman (1993) find that past winners outperform past losers, supporting the profitability of a momentum strategy. As subsequent research confirms the profitability of different trading strategies over varying investment horizons, related studies are naturally expanded to international stock markets, including South Korea. However, the sources of gains in trading strategies are still under debate.

The literature on the sources of gains from trading strategies can be summarized along two lines of research: investor irrationality and systematic risk. Investor irrationality results in overreactions and underreactions. DeBondt and Thaler (1985) argue that an overreaction in the stock market prompts return reversals, which become a source of profits from the contrarian trading strategy. Jegadeesh (1990) illustrates evidence of shorter-term return reversals. Jegadeesh and Titman (1993), Liu *et al.* (1999), and Chan *et al.* (1996) illustrate that performance from the momentum strategy is the result of an underreaction to new stock market information. Using a model based on investor psychology, Barberis *et al.* (1998) examine the manner in which investors form expectations of future profits. They find evidence that stock prices underreact to weak information that is statistically significant, while stock prices overreact to strong information that is statistically insignificant.

With respect to systematic risk, the Capital Asset Pricing Model (CAPM), developed by Sharpe (1964) and others, suggests that expected stock returns are positively and linearly related to the market beta and that the market beta explains differences in cross-sectional stock returns. However, subsequent empirical studies mitigate the CAPM's explanation. Fama and French (1993) recommend a multi-factor model after confirming common factors determining the prices of common stocks and bonds. They argue that beta, firm size, and book-to-market ratio can explain time-series behavior and cross-sectional differences in stock returns. While Fama and French (1996) confirm that the three-factor model explains the long-term reversals suggested by DeBondt and Thaler (1985), the momentum effect is still not explained by the three-factor model.

Arena *et al.* (2008) show that momentum profits increase with idiosyncratic volatility (IVol) in the U.S., suggesting that momentum profits can be realized because of investors' underreactions to firm-specific information as well. They find that portfolios with higher IVol deliver higher momentum profits than those with lower IVol. They determine that this finding is robust after controlling for variables that could affect momentum profits, such as firm size, trading volume, and market risk. Recently, Park and Kim (2011) examine momentum profits in several countries, including South Korea, and find no momentum effects in the Korean stock market for data from 1990 through 2010. Recognizing significant changes in the Korean stock market during the 2000s, Kang *et al.* (2011) find that the momentum effects are confined to large firms. However, their portfolios, in computing momentum profits, are not common in relation to portfolio formation, because they formed portfolios using the top 30% and bottom 30% in stock price performance.

We investigate the profitability of trading strategies in the Korean stock market and analyze the profitability relative to IVol. To accomplish our goal, we first examine the profitability of trading strategies for data from 1997-2009, reflecting the recent movements in the Korean stock market. The extant South Korean literature dealing with the profitability of trading strategies reflects data through 2000. However, the Korean stock market has grown significantly since 2000, and its representative index, known as KOSPI, has been included in stock market indices in developed countries. In addition, the Korean stock market is considered a mature financial market, overcoming internal and external shocks. Furthermore, South Korea is found to be the first country to recover from the global financial crisis of 2008.

Another step to accomplish our goal is to explain the sources of gains from the momentum trading strategy with IVol. While studies in South Korea have confirmed the profitability of the momentum, or the contrarian, strategy and have attempted to explain the sources of gains in several aspects, there is no study analyzing the sources of the gains with the momentum strategy relative to IVol, to our knowledge. Consequently, we examine the impact of the IVol on the profitability of the momentum strategy and explain the results in the behavioral finance literature.

This study contributes to our understanding of the sources of momentum gains in one of the newly matured financial markets. Investigating profitability in trading strategies for sample firms in South Korea during 1997-2009, we confirm that there are excess returns from the momentum strategy, as opposed to the contrarian strategy. A subsequent analysis illustrates that stocks with high IVol generate higher momentum profits than those with stocks with low IVol. We also find that the results are not affected by control factors, such as firm size, share price, and market beta. The time-series analysis confirms that momentum profits are positively related to IVol.

Our results are generally consistent with the findings of Arena *et al.* (2008), who show that momentum returns are higher for high IVol stocks in the U.S. Since the momentum profits are closely related to IVol in South Korea, we also find that the underreaction to firm-specific information is a source of gains in South Korea and the U.S. IVol is a proxy for firm-specific information, but firm-specific information has not been considered in asset pricing. Combined together with those of Arena *et al.* (2008), our results explain why momentum profits persist and increase long after the period studied by Jegadeesh and Titman (1993).

While the positive relationship between momentum profits and IVol in Arena *et al.* (2008) remains consistent with our study, we differ on several points from other studies of investment strategies that are receiving attention in South Korea. Although studies on these trading strategies generally show no momentum profits in South Korea, the sample periods do not include data beyond the 1990s.⁴

The rest of the paper is organized as follows. Section 2 presents a review of the extant literature. Section 3 explains the data, defines the variables, and illustrates the method of analysis. We show and interpret the results from the analysis of investment strategies and sources of momentum returns in Section 4. The results from the time-series regression analysis and the

⁴ See Kim (2000), Lee and Ahn (2002), and Ahn and Lee (2004).

relationship between aggregate IVol and momentum profits are discussed in Section 5. Section 6 summarizes and concludes the study.

2. Literature Review

DeBondt and Thaler (1985, 1988) were the first authors to publish results illustrating that stocks that outperform the market for a period of 3-5 years will consistently underperform after the 3-5 year period is over. Observing the mean reversion of common stocks, they suggest that the effectiveness of the contrarian strategy is based on return reversals. Subsequently, Jegadeesh (1990) reports that excess returns from the contrarian strategy are significant over the short term in the U.S. stock market. However, forming two portfolios for past losers and past winners based on 3-6 months of past return data in the U.S., Jegadeesh and Titman (1993) periodically review accumulated returns of past losers for the following 3-12 months and compare that with a portfolio of past winners, which is found to outperform that of past losers, supporting the momentum strategy.

The sources of gains in momentum or contrarian effects can be divided into two strands. One strand is investor irrationality, resulting in underreactions or overreactions; the other strand is systematic risk, resulting in contribution to stock returns. Debondt and Thaler (1985) argue that an overreaction by investors causes return reversals and allows for excess returns from the contrarian strategy. Daniel *et al.* (1998) explain excess returns in the short term with persistent overreactions by informed traders, who have overconfidence in private information, with subsequent return reversals by corrections from the overreaction. Conversely, Conrad and Kaul (1998) change the time periods in forming and holding portfolios and analyze the performance of momentum and contrarian strategies. They find that the contrarian strategy outperforms in the short and the long term, while the momentum strategy outperforms in the medium term. Lee and Swaminathan (2000) confirm both return momentums in the short term and return reversals in the long term.

For underreactions, Jagadeesh and Titman (1993) initially report excess returns from the momentum strategy. Barberis *et al.* (1998) adopt a model with investor psychology and examine the process of forming future profit expectations. They find that investors underreact to information that is weakly intensive, but statistically significant, while they overreact to information that is highly intensive, but statistically insignificant. Liu *et al.* (1999) and Chan *et al.* (1996) show that the market's underreaction to new information drives up excess returns from the momentum strategy. Hong and Stein (1999) analyze two types of investors, news observers and momentum traders, with their rational constraints, and find that there are underreactions in the short term, while there are overreactions in the long term. These authors suggest that the initial underreaction by news observers to firm-specific information provides profitable opportunities to momentum traders who persistently participate in the market, resulting in overreactions.

Aside from investor irrationality, it has been suggested that systematic risk is a source of excess returns from the momentum or contrarian strategy. Based on the Capital Asset Pricing Model (CAPM) by Sharpe (1964), expected returns are a positive linear function of the market

beta. In addition, cross-sectional differences in expected returns can be explained by the market beta. However, empirical studies illustrate that, in addition to the systematic beta, firm size and book-to-market ratio also affect stock returns, weakening the principle of CAPM. These studies open the door to additional factors to explain the cross-sectional difference in expected returns.

Fama and French (1993) recommend using a multi-factor model to confirm common risk factors in stocks and bonds. They argue that the three stock market factors (e.g., market risk, firm size, and book-to-market ratio) can explain the time-series behavior of, and cross-sectional difference in, expected returns. Fama and French (1996) empirically show that the long-term return reversal identified by De Bondt and Thaler (1985) can be explained by the three-factor model. Lewellen (1999) examines time-series relationships in expected returns, risk, and book-to-market ratio and argues that the book-to-market ratio can be used to both economically and statistically predict expected stock returns.

The related literature is also active outside the U.S. Rouwenhorst (1998) examines the profitability of portfolios by past losers and past winners in 12 countries in Europe and finds that the effectiveness of the momentum strategy is significant in both emerging and developed capital markets. Liu *et al.* (1999) and Ryan and Overmeyer (2004) apply the momentum approach by Jegadeesh and Titman (1993) to England and Germany and confirm the existence of excess returns from the momentum strategy. On the other hand, Mun *et al.* (1999) illustrate the profitability of the contrarian strategy in the French and German stock markets. They also find that the performance of the contrarian strategy is significant in the short term alone, but declines over time. Da Costa (1994) and Bowman and Iverson (1998) confirm return reversals in Brazil and New Zealand, respectively. More recently, Antoniou *et al.* (2006) find the profitability of the contrarian strategy in the short term in the English stock market.

The literature also includes research on the Korean stock market. Most studies on these trading strategies in South Korea do not observe momentum profits for the sample period through the 1990s (Kim, 2000; Lee and Ahn, 2002; Ahn and Lee, 2004). More recently, Park and Kim (2011) examine momentum profits in several countries, including South Korea, and find no momentum effects in the Korean stock market. Using a 6 month time period for forming and holding portfolios, they find a momentum profit of -0.40% with a *t*-statistic of -1.63. One reason that the *t*-statistic is not significant might be that the sample period includes data from both the 1990s and 2000s.

Kang *et al.* (2011) recognize significant changes in the Korean stock market during the 2000s and investigate the momentum effect during that period. They find that the momentum effects are confined to large firms due to significant participation in the Korean stock market by foreign investors in the 2000s. However, their method of computing momentum profits is different from the usual approach. Their winner portfolio comes from the top 30% of firms and their loser portfolio comes from the bottom 30% of firms, as opposed to from the top 10% and bottom 10%, respectively. Because these authors widen the range of firms in the winner and loser portfolios, it is difficult to detect momentum profits.

More recently, Arena *et al.* (2008) find that there is a positive relationship between the profitability of momentum strategies and IVol. In addition, they argue that the underreaction to

firm-specific information is a source of momentum profits. They show that stocks with high IVol generate momentum returns that are higher than those with low IVol. They also confirm that the results are consistent, even after they control for factors such as firm size, trading volume, and systematic risk, which could affect the momentum returns.

IVol has also attracted attention in incomplete markets, where investors cannot completely diversify their investments and are naturally exposed to the remaining risk. Merton (1987) argues that idiosyncratic risk increases expected stock returns, while investors incur expenses to obtain information and participate in the stock market. On the other hand, Shleifer and Vishny (1997) suggest that IVol does not increase expected returns when there are noise traders, because the existence of such traders helps IVol restrict arbitrage trading. Therefore, stocks with high IVol are overpriced and their expected returns are low. Campbell *et al.* (2001) argue that idiosyncratic risk is important in several ways. On the one hand, portfolios held by individual investors might be undiversified, because of holding a large number of individual stocks. On the other hand, the degree of diversification with 20-30 stocks in a portfolio, which is commonly considered well diversified, is still subject to the idiosyncratic risk of individual stocks. Furthermore, by trading stocks to exploit the mispricing of individual stocks, arbitrageurs cannot avoid risk related to IVol, as implied by Shleifer and Vishny (1997). Ang *et al.* (2006) confirm that stocks with high IVol deliver low expected returns. However, Fu (2009) finds significantly positive relationships between conditional IVol and expected returns, supporting Merton's argument.

3. Methodology

3.1 The Data

This study uses stock price, shares outstanding, trading volume, and data from financial statements for common stocks traded on the Korea Exchange (KRX) between 1997 and 2009 with the following criteria: (1) firms belong to industries other than financial services, because firms in financial services are characteristically different from industrial firms; (2) market capitalization, which is the stock price multiplied by the outstanding shares, should be above 10% in the KRX size decile; in addition, their stock prices should be above KRW (Korean Won) 5,000, because stocks with capitalization below 10%, or with prices below KRW 5,000, can distort the results due to liquidity concerns (Jegadeesh and Titman, 2001); (3) the book value of equity is positive, because a negative value makes the book-to-market ratio meaningless; and (4) firms have been listed on the KRX for over 12 months at portfolio formation.

This study also uses some macroeconomic variables and stock market indices. Data are extracted from DataGuide at FnGuide, KIS-Value at Korea Credit Analysis Information, TS2000 at the Association of Listed Corporations, and ECOS (Economic Statistics System) at the Bank of Korea.

3.2 Return Analysis of Investment Strategies

Adopting the approach in Jegadeesh and Titman (1993), we compute and rank the past 6 months of returns for all stocks in the sample at the beginning of each month and use the

rankings to group the stocks into ten equally weighted portfolios. We then hold each portfolio for 3, 6, 9, and 12 months (K : holding period) after the portfolio is formed. We use overlapping portfolios. The number of reported returns in each month is equal to the number of portfolio formation periods. We remove the oldest portfolio at the beginning of each month and add a newly formed portfolio. We then compute the momentum return as the difference ($P10 - P1$) between the return of the top decile (winner: $P10$) and that of the bottom decile (loser: $P1$). The contrarian return is computed as the difference with a negative sign ($P1 - P10$).

When we form a portfolio at the beginning of month t , we measure its return for the past J periods as:

$$PM_t^J = \left(\prod_{j=1}^J (1 + r_{t-j}) \right)^{1/J} - 1, \quad J = 3, 6, 9, 12, \quad (1)$$

where r_t is returns in month t . Following Jagadeesh and Titman (1993), we take overlapping portfolios for holding periods and compute portfolio returns for each holding period ($K = 3, 6, 9, 12$), starting in the month portfolios are formed. For example, we measure the return of a portfolio with K holding periods in month t by averaging the portfolios previously held for K periods up to a portfolio formed at the beginning of month t as:

$$R_{p,t}(K) = \frac{1}{K} \sum_{i=t-(K-1)}^t r_{p_i,t}(K), \quad K = 3, 6, 9, 12, \quad (2)$$

where $r_{p_i,t}(K)$ is a return in month t for a portfolio formed at the beginning of month i with the holding period K . As we compute the portfolio returns in Eq. (2) every month for the sample periods and N numbers of $R_{p,t}(K)$ in each holding period, we obtain a portfolio return with a K holding period by averaging out $R_{p,t}(K)$ as:

$$R_p(K) = \frac{1}{N} \sum_{t=1}^N R_{p,t}(K), \quad K = 3, 6, 9, 12. \quad (3)$$

Finally, a momentum return is computed as the difference between the return of the winner portfolio and that of the loser portfolio:

$$R_S(K) = \frac{1}{N} \sum_{t=1}^N (R_{W,t}(K) - R_{L,t}(K)), \quad K = 3, 6, 9, 12. \quad (4)$$

3.3. Sources of Momentum Returns

3.3.1 Relationship to Systematic Risk

To examine whether the systematic risk can explain momentum returns, we apply the time-series behavior model by Fama and French (1993) and Black, Jensen, and Scholes (1972) to momentum returns:

$$r_{p,t} - r_{f,t} = a_p + b_p (r_{m,t} - r_{f,t}) + s_p SMB_t + h_p HML_t + \varepsilon_{p,t}, \quad (5)$$

where $r_{p,t}$ is a momentum return for the IVol portfolio p in month t , $r_{m,t}$ is the market return, and $r_{f,t}$ is the risk-free rate of return. SMB is the firm size difference as the size of large firms subtracted from that of small firms. HML is the difference in book-to-market ratios (B/M) as the low ratio subtracted from the high ratio. This regression is useful in examining how the three factors of market risk, firm size, and book-to-market ratio explain stock returns.

We then rearrange the data to extract a risk factor in the three-factor model. At the end of March for every year in the dataset, we divide the sample into 2 groups by size: small and large firms. Ignoring firm size, we then divide the sample into three groups based on their book-to-market ratio: top 30%, middle 40%, and bottom 30%. Combining the book-to-market ratio with the firm size, we form six portfolios. These portfolios are reformed at the end of each year. Assuming the investment weight in each portfolio remains the same, we compute the monthly returns for 12 months from the portfolio formation time.

The SMB portfolio is a proxy for a risk factor in stock returns related to firm size and is computed by subtracting the means of the monthly stock returns on the three large-firm portfolios from the means of monthly stock returns on the three small-firm portfolios. While portfolio returns on small firms and on large firms are influenced by book-to-market ratios, the SMB portfolio removes the influence from B/M by taking the difference between the returns of small firms and those of large firms and focusing on the return differences by their firm size.

Similarly, the HML portfolio is a proxy for a risk factor in stock returns related to B/M and is computed by subtracting the means of monthly returns on the two portfolios with a low B/M from the means of the monthly returns on the two portfolios with a high B/M . Hence, the HML portfolio removes the influence from the firm size by taking the difference between returns of low B/M firms and those of high B/M firms and focusing on the return differences by B/M . The proxy for a market factor is obtained by subtracting risk-free rates from returns on the Korea Composite Stock Price Index (KOSPI).

We also include the other variables in the Fama and French (1993) three-factor model. For risk-free rates of return, we use daily returns on Monetary Stabilization Bonds (MSB) compounded to monthly returns. Monthly returns on KOSPI are used as the market returns. The firm size is computed from the outstanding shares of common stock multiplied by the closing price.

3.3.2 Relationship between IVol and Returns from Investment Strategies

Following Ali *et al.* (2003), Wurgler and Zhuravskaya (2002), and Arena *et al.* (2008), we compute IVol using the following model:

$$r_{i,t} = \alpha_i + b_{0,i}r_{m,t} + b_{1,i}r_{m,t-1} + \varepsilon_{i,t}, \quad (6)$$

where $r_{i,t}$ is a return on stock i on date t , and $r_{m,t}$ and $r_{m,t-1}$ are the market returns on date t and $t-1$, respectively. The lagged term $r_{m,t-1}$ is added to reflect the effects of potential non-synchronous trading (Dimson, 1979). We estimate Eq. (6) using the data for the past 12 months when we form portfolios at the end of each month based on their momentum returns. IVol is obtained from the standard deviation of residuals, $\varepsilon_{i,t}$.

3.4 Time-Series Regression Analysis on Aggregate Idiosyncratic Volatility

We examine the time-series relationship between the aggregate IVol and momentum profits and the cross-sectional analysis. While there is no theory or empirical evidence that implies a horizon at which the momentum should be affected by momentum returns, Arena et al. (2008) conduct an analysis at an annual frequency, because they use a large sample. We perform the examination quarterly to preserve our sample size and maintain a balance between statistical significance and some noise in the high frequency returns.

Cooper *et al.* (2004) illustrate that momentum profits are related to lagged one-year market returns; we control for this factor. We are also concerned about possible autocorrelation, and thus, control for the lagged momentum return in our regressions. To remove influences from macroeconomic factors in our results, we control for dividend yield, default spread, and term spread. Hence, we estimate the following regression:

$$\begin{aligned} \text{MomRet}_t = & a + b_1 \text{AggIVol}_{t-1} + b_2 \text{MktVol}_{t-1} + b_3 \text{TotVol}_{t-1} + b_3 \text{MomRet}_{t-1} \\ & + b_4 1\text{YMktRet}_{t-1} + b_5 \text{DP}_{t-1} + b_6 \text{Term}_{t-1} + b_7 \text{Def}_{t-1} + b_8 \text{MSB1Y}_{t-1} + \varepsilon_t, \end{aligned} \quad (7)$$

where MomRet_t is the cumulative return to the momentum strategy over each quarter. We compute momentum profits for each month as the difference in returns between past losers and past winners. The momentum profits accumulated across all months for each quarter.

MktVol_{t-1} represents the market volatility over the past year, TotVol_{t-1} is the average total stock volatility over the past year, 1YMktRet_{t-1} is the one-year market return, DP_{t-1} is the lagged dividend yield, Term_{t-1} is the lagged term spread, Def_{t-1} is the lagged default spread, and MSB1Y_{t-1} is the lagged one-year Monetary Stabilization Bonds (MSB). The default spread is the difference between AA corporate bond yields and ten-year T-bond yields. We calculate the term spread as the difference between ten-year T-bond yields and one-year MSB rates.

AggIVol_{t-1} is the average aggregate IVol over the previous year. The IVol is estimated each month for each firm using the Fama and French (1993) 3-factor model:

$$\begin{aligned} r_{i,t} - r_{f,t} = & a_i + b_{1i}(r_{m,t} - r_{f,t}) + b_{2i}(r_{m,t-1} - r_{f,t-1}) + s_{1i} \text{SMB}_t \\ & + s_{2i} \text{SMB}_{t-1} + h_{1i} \text{HML}_t + h_{2i} \text{HML}_{t-1} + \varepsilon_{i,t}, \end{aligned} \quad (8)$$

where r_i is the daily return on security i , r_f is the risk-free rate, r_m is the return on KOSPI, and ε_i is the regression residual for the IVol. We take an equal-weighted average of the IVol across all

sample firms to compute the aggregate IVol. Finally, we calculate the average aggregate IVol across all months in a year.

4. Results from the Empirical Analysis

4.1 Momentum Portfolios and the Performance of Investment Strategies

Table 1 illustrates the performance of the momentum portfolios by their holding periods when the momentum period is fixed at 6 months. Examining the returns of the ten portfolios, we find that the monthly returns take on the pattern of a U-shape by decreasing and then increasing in the stock price momentum regardless of their holding periods.

When we compare the returns of the loser portfolios (P1) with those of the winner portfolios (P10), the winner portfolios outperform the loser portfolios across all of the holding periods. Hence, the momentum strategy of buying the winner and selling the loser portfolio should deliver excess returns. More specifically, the momentum returns are 9.39%, 2.77%, 1.73%, and 1.41%, corresponding to their holding periods of 3, 6, 9, and 12 months, respectively. All of the t -statistics related to those returns are statistically significant. While momentum profits decrease with holding periods, we focus on the 6 month holding period to make the results comparable with those of Arena *et al.* (2008).

While the momentum portfolios and their performances are evaluated for the entire sample period, these results can change when we examine sub-periods, as shown in Jegadeesh and Titman (1993). Hence, we divide the sample into two sub-periods. During the two sub-periods, 1997-2002 and 2003-2009, in the Korean stock market, the KOSPI repeatedly moved up and down and maintained its level below 1,000 until 2002, after bottoming out in 1998 from the effect of the IMF bailouts. While KOSPI increased during 2003-2007, it turned downward due to the financial crisis in 2008 and then moved upward afterwards. We divided the sub-periods around 2002 because of distinct market trends. There is no clear trend of up or down in the Korean stock market during the first sub-period, but the second sub-period is recognized as a bull market, in general, except for a temporary bear market just after the financial crisis in 2008.

Reflecting distinct characteristics, Table 2 exhibits the results from the two sub-periods of 1997-2002 and 2003-2009. The winner portfolios (P10) provide higher returns than do the loser portfolios (P1) in both sub-periods. Accordingly, momentum returns (P10-P1) for the first and the second period are statistically significant with 2% and 3.42%, respectively. This result supports momentum strategies in realizing economically significant excess returns, unlike the findings by the extant studies on the Korean stock market.

Chui *et al.* (2012) find the momentum profit in Korea to be -0.0039% for the data period of 1985-2003; this is not significant with a t -statistic of -0.81. Chui *et al.* (2000) use data from 1978-2000 and find the momentum profit to be -0.107%, which is also not significant, with a t -statistic of -0.23. When these authors examine the sub-period of 1997-2000, they find the momentum profit to be 0.268%. Although the momentum profit is still not significant at a t -

statistic of 0.09, it appears that the stock market dynamics in Korea have changed around 1997, which is the beginning year in our sample.

One explanation for this phenomenon could be the trend in changes in the Korean stock market. While the extant research studies examine data in the 1990s, the economy in South Korea significantly developed, relative to the world economy, in the 2000s, which is the time period prevailing in this study. Another explanation could be a difference in methodology. Conrad and Kaul (1998) argue that the performance of investment strategies changes, depending on how one sets the momentum and holding periods.

4.2 Profitability of Investment Strategies and Idiosyncratic Volatility

4.2.1 Momentum Profitability Based on Idiosyncratic Volatility

Arena *et al.* (2008) find that the profitability of momentum strategies is positively related to IVol. They argue that the underreaction to firm-specific information is a source of gains in momentum returns. Based on the methodology of Arena *et al.* (2008), we examine the performance of portfolios independently classified by IVol and price momentum. More specifically, sorting the sample stocks by IVol, we divide the sample into two groups: IVol 1 (low) and IVol 2 (high). Returning to the entire sample, we sort the sample by price momentum and create ten portfolios, from P1 (loser) to P10 (winner). The two independent sorts provide portfolios based on IVol and price momentum.

Table 3 summarizes the profitability of portfolios constructed by price momentum and IVol. While both momentum returns are positive and statistically significant, that of the high IVol portfolio (IVol 2) is higher at 3.91% (*t*-value: 3.38) than that of the low IVol portfolio (IVol 1) at 1.96% (*t*-value: 5.70). The difference in momentum returns between the two IVol portfolios is economically significant at 1.95% per month and is statistically different from zero at the 10% significance level (*t*-value: 1.72). While the higher momentum return of the high IVol portfolio is driven by the underperformance of high IVol losers in Arena *et al.* (2008), the return is driven by the outperformance of high IVol winners in this table. Both losers and winners in the high IVol portfolio deliver returns at 3.33% and 7.24%, which are higher than those in the low IVol portfolio at 0.82% and 2.78%, respectively.

Although Arena *et al.* (2008) do not explicitly investigate the relationship between IVol and expected stock returns, they claim that the results are consistent with those of Ang *et al.* (2006), by pointing out that high IVol stocks have lower returns than those of low IVol stocks. However, a closer examination of Table 3 in Arena *et al.* (2008) shows otherwise. When we compare the returns of low IVol with those of medium IVol in their Table 3, we find that the returns of medium IVol are generally higher than those of low IVol. It appears that while returns of the low IVol portfolio are lower than those of the medium IVol portfolio, they are higher than those of the high IVol portfolio. This observation also opens the door to the debate about the relationship between IVol and expected returns.

Merton (1987) argues that firms with larger firm-specific variances deliver higher average returns to compensate for holding less diversified portfolios. The argument implies that firms

with higher IVol should generate higher average returns. While Ang *et al.* negate the theory by finding that firms with higher IVol actually deliver lower average returns, Fu (2009) illustrates that the relationship is positive by recognizing that IVol is time-varying. However, the results in our Table 3 support the argument by Merton, because the higher momentum return of the high IVol portfolio is driven by the outperformance of the high IVol winners, as opposed to the findings in Arena *et al.* (2008).

4.2.2 Momentum Profitability and Systematic Risk

We examine the momentum profitability (Table 3) relative to systematic risk to determine whether the momentum profitability can be explained by systematic risk. Table 4 illustrates the results from running the three-factor model by Fama and French (1993). Both momentum profits with IVol 1 and IVol 2 illustrate that intercepts are positive at the 1% significance level. In addition, the difference in momentum profits between the portfolios of IVol 1 and IVol 2 is found to have a positive intercept, which implies that the impact of IVol on momentum profits cannot be explained by the Fama and French (1993) model. These results are consistent with the findings in Arena *et al.* (2008).

Recall that momentum profits are related to IVol for several reasons. The first reason is that IVol is considered to be a proxy for firm-specific information. Jagadeesh and Titman (1993) and Barberis *et al.* (1998) argue that momentum profits result from underreactions to firm-specific information. Thus, stocks with high IVol deliver higher momentum profits, according to behavioral finance.

The second reason is that IVol discourages arbitrage trading. Shleifer and Vishny (1997) suggest that when there are noise traders, investors have difficulty rationalizing stock pricing with higher IVol and are discouraged from arbitrage trading by IVol. This finding implies that when stocks have high IVol, even with significant momentum profits, the momentum profits can be persistent in the absence of arbitrage trading. Hence, the higher the IVol in stocks, the higher the momentum profit.

4.3 Robustness Tests

The finance literature shows that momentum profits are influenced by several factors, including firm size, transaction costs, and turnovers. Jegadeesh and Titman (1993) and Hong *et al.* (2000) find that momentum profits are high in small firms. This finding implies that the impact of IVol on momentum profits can be reduced when we control for firm size. Alternatively, Shleifer and Vishny (1997) argue that transaction costs also discourage arbitrage trading. As IVol and transaction costs share common characteristics, it is natural to expect that transaction costs influence momentum profits. Analyzing the relationship between momentum profits and turnover, Lee and Swaminathan (2000) illustrate that turnover is positively related to momentum profits. The results in Arena *et al.* (2008) generally confirm these findings in the literature.

Considering the findings in the literature, we control for the factors affecting momentum profits, such as firm size, transaction costs, and turnover, to examine whether the impact of IVol on momentum profits is still significant. Firm size is defined as market capitalization, or the

stock price multiplied by the outstanding shares. Transaction costs can be inferred from share price, because they are considered to have an inverse relationship with share price (Stoll, 2000). Turnover is annualized for the portfolio stocks over the year prior to portfolio formation.

The results of momentum size are exhibited in Table 5 by IVol, firm size, stock price, and turnover. IVol is found to take on a U-shape, where IVols at both ends of the price momentum are high (P1: 4.47% and P10: 5.91%) and reduced as price momentum moves toward the middle portfolio (P5: 3.61%). Turnover shows a similar pattern to IVol; the price momentum increases with firm size and stock price. While the patterns related to IVol, price, and turnover are consistent with the findings in Arena *et al.* (2008), the pattern related to firm size is not. The pattern that the price momentum increases with stock price appears to be unique to the Korean stock market, which has been established as one of the developed countries.

The finding by Jagadeesh and Titman (1993) and Hong *et al.* (2000), that momentum profits are higher among smaller stocks, prompts concerns that the impact of IVol on momentum profits can be reduced by size effects. The concern is cleared in Arena *et al.* (2008), finding that momentum returns are still affected by IVol after controlling for the size effects. Table 6 illustrates the monthly returns of portfolios formed by price momentum, IVol, and firm size. However, we cannot confirm the finding by Arena *et al.* (2008) due to the small sample size.

Arena *et al.* (2008) use a data sample for the years of 1965-2002; our data, however, are restricted to the period of 1997-2009 in the Korean stock market. When we examine the performance of momentum strategies by different IVol in each portfolio of various firm sizes, the difference is not found to be significant. The portfolio for small firms has a return of 1.09% ($t = 1.117$) and for large firms, a return of 0.36% ($t = 0.3873$). That is, the difference in momentum profits between portfolios with low and high IVol is not significant in small or large firms. Although our results are not statistically significant, our results are consistent with the literature, by illustrating the pattern that the momentum return of small firms is weakly higher than that of large firms.

While stock prices are inversely related to transactions costs, as presented in Stoll (2000), Shleifer and Vishny (1997) suggest that arbitrage is restricted by transactions costs. Considering the price effects, Arena *et al.* (2008) confirm that the effect of IVol on momentum returns remains after controlling for the price effects. We independently sort the portfolios on IVol, stock price, and realized returns. Table 7 exhibits the monthly returns of portfolios formed by the price momentum, IVol, and stock price. We find that differences in momentum returns between low and high IVol portfolios in each stock price (low or high) are 1.83% ($t = 1.67$) and 1.28% ($t = 1.19$), respectively. Although the differences are not statistically significant, due to the small sample size, we find that the impact of IVol on momentum profits in low-priced stocks is similar to that of high-priced stocks. Thus, our results are consistent with those in Arena *et al.* (2008).

Recognizing the findings of Lee and Swaminathan (2000), that momentum returns tend to increase in turnover, Arena *et al.* (2008) perform independent sorts on IVol, turnover, and realized returns. Arena *et al.* (2008) report that the differences in momentum returns between high and low IVol portfolios are significant for both low and medium turnover stocks, but not for high turnover stocks. That is, Arena *et al.* (2008) suggest that IVol still increases the explanatory

power on momentum returns after controlling for turnover. We examine a similar relationship to Arena et al. (2008) and provide the results in Table 8. While Arena et al. (2008) cannot find a significant difference in momentum profits between high and low IVol portfolios for high turnover stocks, our results illustrate that the difference at 1.73 ($t = 1.30$) for high turnover stocks is approximately the same as that at 1.69 ($t = 1.33$) for low turnover stocks. Although both are not statistically significant, we find that our results are consistent with the implication by Lee and Swaminathan (2000). We also observe that the price momentum is high among portfolios with a high IVol.

5. Relationship between Aggregate Idiosyncratic Volatility and Momentum Profits

5.1 Time-Series Regression Analysis

Since the results so far reflect on the cross sectional relationship between IVol and momentum profits, we now examine the time-series relationship between aggregate IVol and momentum profits. Table 9 shows insignificant evidence that momentum profits are positively related to lagged one-year market returns. While Arena et al. (2008) confirm the implication by Cooper *et al.* (2004), with lagged three-year returns using U.S. data, we cannot confirm this implication with lagged one-year market returns for the Korean stock market. The coefficients on lagged momentum returns are insignificant in all of the regressions, indicating that momentum returns are not autocorrelated at the quarterly frequency. However, all of the regressions exhibit a positive relationship between lagged IVol and momentum returns, except for the insignificant first regression.

While Arena et al. (2008) suggest that all the regressions illustrate positive and significant relationship, the first regression shows that the relationship is not significant at the 5% level (t -statistic of 1.77). In our Table 9, the significant coefficients on AggIVol_{t-1} take on values from 5.182 to 11.955, with different control variables, and t -statistics ranging from 2.12 to 3.27, indicating statistical significance at the 1% to 5% levels. When we control for market volatility and total volatility, we still find a significant and positive relationship between momentum profits and aggregate IVol.

Summarizing the results, we illustrate evidence that IVol is positively related to momentum profits in the data from South Korea. The evidence is consistent with the results in Arena *et al.* (2008) for U.S. data. Although our results are not as strong as that in Arena et al. (2008), due to data obstacles in South Korea, the result is robust for the variety of control variables.

5.2 The Persistence of Momentum Effect by the Idiosyncratic Volatility Trend

While some anomalies in the stock market have been explained in the literature, the momentum effect has not. More specifically, Schwert (2003) shows that common anomalies by the small-firm effect and the value effect disappear following the sample periods initially identified with these anomalies. However, the momentum anomaly remains.

Momentum profits are not only persistent, they also increase (Jegadeesh and Titman, 1993). We report (Table 2) that the average momentum return is 2.00% for 1997-2002 and 3.42% for 2002-2009. We suggest that IVol is a significant reason to explain why momentum profits are persistent and increase over time, even in the Korean stock market. Our results in Table 9 imply a positive time-series relationship between momentum returns and IVol.

Arena *et al.* (2008) also illustrate that IVol displays an upward trend over the period of 1965-2002 for U.S. data. Combined with the findings by Arena *et al.* (2008), our results suggest that the long-term rise in IVol likely drives the increase in momentum profits, following Jegadeesh and Titman (1993), for both U.S. and South Korean data.

6. Conclusions

We examine the relationship between momentum profits and IVol, where IVol is a variable that has developed increasing interest in the momentum trading literature in the Korean stock market. We confirm momentum profits and find that IVol plays an important role in momentum profits. We show that stocks with higher IVol exhibit momentum returns larger than stocks with lower IVol. Our results are robust after we control for factors such as firm size, share price, and market beta. The relationship between momentum returns and IVol is statistically significant and large, which is consistent with Arena *et al.* (2008). However, the relationship is primarily driven by high IVol winners.

Our findings support the view that momentum profits are driven by underreactions to firm-specific information, which is implied by IVol. The results also support the hypothesis that IVol poses as an important limit to arbitrage. Momentum returns are largest in the portfolios with the highest IVol, exhibiting a momentum effect that is more readily arbitrated away to stocks with low firm-specific information. Our results are also consistent with the asymmetric-information model in Wang (1993).

The time-series evidence illustrates a positive relationship between momentum returns and aggregate IVol. This result complements the cross-sectional analysis, supporting our view that the momentum effect can be explained by aggregate IVol. The result also explains the persistence and increase in momentum returns in the late 1990s and 2000s. While common anomalies disappear following the sample period reviewed by the initial studies, momentum returns increase following the sample period of Jegadeesh and Titman (1993). We suggest that our finding of a positive time-series relationship between IVol and momentum returns is consistent with the long-term increase in firm-specific volatility observed by Campbell *et al.* (2001) and provides a partial explanation for momentum returns.

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Table 1: Monthly returns for portfolios based on price momentum and holding periods

The sample includes common stocks traded on the Korea Exchange (KRX) between 1997 and 2009 with the following criteria: (1) firms belong to industries other than the finance industry; (2) market capitalization should be above 10% and stock prices above KRW 5,000; (3) the book value of equity is positive; and (4) firms have enough data for an analysis. Momentum portfolios are formed based on the past six months of returns and held for K (=3, 6, 9, 12) months. P1 to P10 exhibit momentum portfolios, with P1 as past losers and P10 as past winners. Returns are shown as percentages. The *t*-statistics are illustrated in parentheses. Asterisks represent significance at the 10% (*), 5% (**), and 1% (***) levels.

	K = 3		K = 6		K = 9		K = 12	
P1 (Loser)	2.64**	(2.03)	2.43**	(2.36)	2.73***	(2.78)	2.66***	(2.77)
P2	0.72	(0.77)	2.22***	(2.59)	2.46***	(2.87)	2.44***	(2.88)
P3	0.74	(0.82)	2.09**	(2.53)	2.29***	(2.73)	2.35***	(2.82)
P4	0.96	(1.03)	1.99**	(2.46)	2.31***	(2.86)	2.35***	(2.90)
P5	1.57*	(1.81)	2.19***	(2.73)	2.27***	(2.84)	2.38***	(2.98)
P6	2.12**	(2.43)	2.24***	(2.81)	2.27***	(2.85)	2.31***	(2.95)
P7	2.59***	(3.00)	2.67***	(3.27)	2.61***	(3.26)	2.65***	(3.33)
P8	3.13***	(3.65)	2.74***	(3.37)	2.55***	(3.18)	2.55***	(3.16)
P9	4.18***	(4.58)	3.15***	(3.94)	2.89***	(3.58)	2.82***	(3.48)
P10 (Winner)	12.0***	(7.99)	5.20***	(5.74)	4.46***	(5.03)	4.07***	(4.55)
P10 – P1	9.39***	(7.22)	2.77***	(4.84)	1.73***	(3.59)	1.41***	(3.04)

Table 2: Monthly returns for portfolios based on price momentum with sub-periods

The sample includes common stocks traded on the Korea Exchange (KRX) between 1997 and 2009 with the following criteria: (1) firms belong to industries other than the finance industry; (2) market capitalization should be above 10% and stock prices above KRW 5,000; (3) the book value of equity is positive; and (4) firms have enough data for an analysis. Momentum portfolios are formed based on the past six months of returns and held for K (=3, 6, 9, 12) months. P1 to P10 exhibit momentum portfolios, with P1 as past losers and P10 as past winners. Returns are shown as percentages. The *t*-statistics are illustrated in parentheses. Asterisks represent significance at the 10% (*), 5% (**), and 1% (***) levels.

	1997-2009		1997-2002		2003-2009	
P1 (Loser)	2.43**	(2.36)	2.62***	(1.37)	2.27**	(2.23)
P2	2.22***	(2.59)	2.18**	(1.41)	2.22**	(2.51)
P3	2.09**	(2.53)	1.62	(1.08)	2.48**	(2.90)
P4	1.99**	(2.46)	1.47	(1.00)	2.43**	(2.93)
P5	2.19***	(2.73)	1.53	(1.05)	2.75***	(3.32)
P6	2.24***	(2.81)	1.80*	(1.22)	2.62***	(3.33)
P7	2.67***	(3.27)	2.13**	(1.45)	3.12***	(3.67)
P8	2.74***	(3.37)	2.22**	(1.50)	3.18***	(3.82)
P9	3.15***	(3.94)	2.53**	(1.82)	3.69***	(4.08)
P10 (Winner)	5.20***	(5.74)	4.60***	(3.05)	5.69***	(5.26)
P10 – P1	2.77***	(4.84)	2.00**	(2.01)	3.42***	(5.37)

Table 3: Monthly returns on portfolios formed by price momentum and IVol

The table presents average monthly percentage returns on portfolios based on independent sorts of price momentum and IVol. The sample includes common stocks traded on the Korea Exchange (KRX) between 1997 and 2009 with the following criteria: (1) firms belong to industries other than the finance industry; (2) market capitalization should be above 10% and stock prices above KRW 5,000; (3) the book value of equity is positive; and (4) firms have enough data for an analysis. Stocks are sorted into two groups (IVol 1: low, IVol 2: high) by IVol. IVol is computed as the standard deviation of the residuals from the regression with data for the past twelve months:

$$r_{i,t} = \alpha_i + b_{0,i}r_{m,t} + b_{1,i}r_{m,t-1} + e_{i,t}, \text{ where } r_{i,t} \text{ is the daily return on security } i \text{ and } r_{m,t} \text{ is the value-weighted KOSPI return.}$$

We construct the momentum for a portfolio from the past six months of returns and hold the portfolios for six months in each group. P1 through P10 exhibit momentum portfolios, with P1 as past losers and P10 as past winners. Returns are shown as percentages. The *t*-statistics are illustrated in parentheses. Asterisks represent significance at the 10% (*), 5% (**), and 1% (***) levels.

	IVol 1 (Low)		IVol 2 (High)		IVol 2 – IVol 1	
P1	0.82	(1.02)	3.33***	(2.82)		
P2	0.66	(0.91)	3.29***	(3.22)		
P3	0.82	(1.16)	3.26***	(3.24)		
P4	1.02	(1.39)	2.77***	(2.82)		
P5	1.10	(1.61)	2.78***	(2.81)		
P6	1.13*	(1.67)	3.62***	(3.52)		
P7	1.59**	(2.30)	3.75***	(3.60)		
P8	1.75**	(2.46)	3.61***	(3.67)		
P9	2.00***	(2.86)	4.68***	(4.71)		
P10	2.78***	(3.80)	7.24***	(5.28)		
P10 – P1	1.96***	(5.70)	3.91***	(3.38)	1.95*	(1.72)

Table 4: Regression with momentum returns using the Fama and French (1993) three-factor model

The table shows regression results of momentum returns using the Fama and French (1993) three-factor model. We use stock price and financial data for common stocks traded on the Korea Exchange (KRX) between 1997 and 2009 with the following criteria: (1) firms belong to industries other than the finance industry; (2) market capitalization should be above 10% and stock prices above KRW 5,000; (3) the book value of equity is positive; and (4) firms have enough data for an analysis. Stocks are sorted into two groups (IVol 1: low, IVol 2: high) by IVol. IVol is computed as the standard deviation of the residuals from the regression using data from the past twelve months:

$$r_{i,t} = \alpha_i + b_{0,i}r_{m,t} + b_{1,i}r_{m,t-1} + e_{i,t}, \text{ where } r_{i,t} \text{ is the daily return on security } i \text{ and } r_{m,t} \text{ is the value-weighted market return.}$$

In the row headings, r_f is the risk-free rate and r_m is the return on the KOSPI market index; SMB and HML are computed using the method in Fama and French (1993). Asterisks represent significance at the 10% (*), 5% (**), and 1% (***) levels.

	Variable	Coefficient	<i>t</i> -statistic	<i>p</i> -value	Adjusted R^2
IVol 1	Alpha	0.0242***	15.63	0.00	0.08
	$r_m - r_f$	0.7093***	23.65	0.00	
	SMB	0.3360***	8.68	0.00	
	HML	0.2873***	6.68	0.00	
IVol 2	Alpha	0.0457***	11.33	0.00	0.09
	$r_m - r_f$	1.1168***	10.26	0.00	
	SMB	0.5207***	3.72	0.00	
	HML	-0.0176	0.11	0.91	
IVol 2 – IVol 1	Alpha	0.0215***	7.30	0.00	0.01
	$r_m - r_f$	0.4075***	3.90	0.00	
	SMB	0.1847	1.38	0.17	
	HML	-0.3049**	2.04	0.04	

Table 5: Descriptive statistics for portfolios with price momentum and stock characteristics

The table presents descriptive statistics for portfolios with price momentum and other stock characteristics. We use stock price, trading volume, and financial data for common stocks traded on the Korea Exchange (KRX) between 1997 and 2009 with the following criteria: (1) firms belong to industries other than the finance industry; (2) market capitalization should be above 10% and stock prices above KRW 5,000; (3) the book value of equity is positive; and (4) firms have enough data for an analysis. IVol is computed as the standard deviation of the residuals from the regression using data from the past twelve months: $r_{i,t} = \alpha_i + b_{0,i}r_{m,t} + b_{1,i}r_{m,t-1} + e_{i,t}$, where $r_{i,t}$ is the daily return on security i and $r_{m,t}$ is the value-weighted KOSPI return. We construct the momentum for a portfolio from the past six months of returns and hold the portfolios for six months in each group. P1 through P10 exhibit momentum portfolios, with P1 as past losers and P10 as past winners. Price means the stock price, which is multiplied by the number of outstanding shares to obtain the size in each portfolio. IVol, size, and price are recorded on the date of portfolio formation. Turnover, in percent, is annualized for the stocks in the portfolio for one year before the portfolio is formed.

	IVol %	Size KRW in 100 million	Price KRW	Turnover %
P1	4.47	7,762.60	25,133.58	17.06
P2	3.77	9,445.40	31,697.61	12.13
P3	3.63	10,471.12	32,056.01	12.00
P4	3.62	10,115.07	35,672.84	11.05
P5	3.61	10,043.03	32,859.11	10.91
P6	3.56	11,540.59	36,649.75	10.90
P7	3.70	12,246.60	40,958.04	12.11
P8	3.72	12,835.65	39,495.50	12.31
P9	3.83	14,598.64	43,913.82	13.72
P10	5.91	12,457.89	40,424.81	25.02

Table 6: Portfolio returns by price momentum, IVol, and firm size

The table shows average monthly percentage returns on portfolios sorted by firm size and IVol. The sample includes common stocks traded on the Korea Exchange (KRX) between 1997 and 2009 with the following criteria: (1) firms belong to industries other than the finance industry; (2) market capitalization should be above 10% and stock prices above KRW 5,000; (3) the book value of equity is positive; and (4) firms have enough data for an analysis. IVol is computed as the standard deviation of the residuals from the regression using data from the past twelve months:

$r_{i,t} = \alpha_i + b_{0,i}r_{m,t} + b_{1,i}r_{m,t-1} + e_{i,t}$, where $r_{i,t}$ is the daily return on security i and $r_{m,t}$ is the value-weighted KOSPI return.

We construct the momentum for a portfolio from the past six months of returns and hold the portfolios for six months in each group. P1 through P10 exhibit momentum portfolios, with P1 as past losers and P10 as past winners. Size is the stock price multiplied by the number of outstanding shares. IVol and size are recorded on the date of portfolio formation. Asterisks represent the significance at the 10% (*), 5% (**), and 1% (***) levels.

		IVol 1 (Low)		IVol 2 (High)		IVol 2 – IVol 1	
Size 1 (Small)	P1	0.05	(0.06)	3.16**	(2.50)		
	P5	0.71	(1.07)	3.38***	(3.14)		
	P10	1.63**	(2.15)	5.83***	(4.80)		
	P10 – P1	1.58***	(3.42)	2.68***	(2.82)	1.09	(1.12)
Size 2 (Large)	P1	1.47*	(1.76)	2.99***	(2.71)		
	P5	1.53**	(2.19)	3.43***	(3.24)		
	P10	3.37***	(4.54)	5.24***	(4.22)		
	P10 – P1	1.90***	(4.38)	2.25**	(2.46)	0.36	(0.39)

Table 7: Portfolio returns by price momentum, IVol, and stock price

The table presents average monthly percentage returns on portfolios sorted by stock price and IVol. The sample includes common stocks traded on the Korea Exchange (KRX) between 1997 and 2009 with the following criteria: (1) firms belong to industries other than the finance industry; (2) market capitalization should be above 10% and stock prices above KRW 5,000; (3) the book value of equity is positive; and (4) firms have enough data for an analysis. IVol is computed as the standard deviation of the residuals from the regression using data from the past twelve months: $r_{i,t} = \alpha_i + b_{0,i}r_{m,t} + b_{1,i}r_{m,t-1} + e_{i,t}$, where $r_{i,t}$ is the daily return on security i and $r_{m,t}$ is the value-weighted KOSPI return. We construct the momentum of a portfolio from the past six months of returns and hold the portfolios for six months in each group. P1 through P10 exhibit momentum portfolios, with P1 as past losers and P10 as past winners. Price means the stock price. IVol and price are recorded on date of portfolio formation. Asterisks represent significance at the 10% (*), 5% (**), and 1% (***) levels.

		IVol 1 (Low)		IVol 2 (High)		IVol 2 – IVol 1	
Price 1 (Low)	P1	0.43	(0.53)	2.74*	(1.90)		
	P5	0.89	(1.08)	2.40**	(2.41)		
	P10	1.76**	(2.13)	5.91***	(5.07)		
	P10 – P1	1.33**	(2.83)	3.17***	(2.90)	1.83*	(1.67)
Price 2 (High)	P1	1.17	(1.48)	2.48**	(2.40)		
	P5	1.30**	(2.04)	2.93***	(3.10)		
	P10	2.97***	(4.18)	5.57***	(4.68)		
	P10 – P1	1.80***	(4.96)	3.08***	(2.91)	1.28	(1.19)

Table 8: Portfolio returns by price momentum, IVol, and turnover

The table shows average monthly percentage returns on portfolios sorted by turnover and IVol. The sample includes common stocks traded on the Korea Exchange (KRX) between 1997 and 2009 with the following criteria: (1) firms belong to industries other than the finance industry; (2) market capitalization should be above 10% and stock prices above KRW 5,000; (3) the book value of equity is positive; and (4) firms have enough data for an analysis. IVol is computed as the standard deviation of the residuals from the regression using data from the past twelve months:

$$r_{i,t} = \alpha_i + b_{0,i}r_{m,t} + b_{1,i}r_{m,t-1} + e_{i,t}, \text{ where } r_{i,t} \text{ is the daily return on security } i \text{ and } r_{m,t} \text{ is the value-weighted KOSPI return.}$$

We construct the momentum of a portfolio from the past six months of returns and hold the portfolios for six months in each group. P1 through P10 exhibit momentum portfolios, with P1 as past losers and P10 as past winners. IVol is recorded on the date of portfolio formation. Turnover in percent is annualized for stocks in the portfolio for one year before the portfolio is formed. Asterisks represent significance at the 10% (*), 5% (**), and 1% (***) levels.

		IVol 1 (Low)		IVol 2 (High)		IVol 2 – IVol 1	
Turnover 1 (Low)	P1	1.20	(1.57)	1.20	(1.25)		
	P5	1.60**	(2.39)	3.22***	(3.34)		
	P10	2.76***	(3.69)	4.54***	(4.07)		
	P10 – P1	1.56***	(3.92)	3.25***	(3.57)	1.69	(1.33)
Turnover 2 (High)	P1	1.17	(1.26)	2.23**	(2.09)		
	P5	2.01***	(2.72)	2.67**	(2.56)		
	P10	3.03***	(3.44)	6.38***	(4.14)		
	P10 – P1	2.43***	(3.56)	4.15***	(3.29)	1.73	(1.30)

Table 9: Regression results with quarterly returns on aggregate IVol

The table presents results of the time-series relationship between aggregate IVol and momentum profits. The dependent variable is the cumulative momentum return (the difference between past winners (P10) and past losers (P1)) across the months of each quarter (MomRet_t). The independent variables are lagged values of MomRet, market volatility (MktVol_{t-1}), average total stock volatility (TotVol_{t-1}), one-year KOSPI value-weighted market return (1YMktRet_{t-1}), dividend yield on the KOSPI (DP_{t-1}), term spread, the difference between five-year Korean T-bond yields and one-year rates of the Monetary Stabilization Bonds (Term_{t-1}), default spread, the difference between AA-corporate bond yields and five-year Korean T-bond yields (Def_{t-1}), and one-year rates of the Monetary Stabilization Bonds (TB1Y_{t-1}). AggIVol_{t-1} is the equal-weighted average across sample stocks of the standard deviation of residuals from the Fama and French (1993) three-factor model using daily returns over the 12 months before the portfolio formation. The t-statistics are shown in parentheses. Asterisks represent significance at the 10% (*), 5% (**), and 1% (***) levels.

	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	0.026 (0.40)	-0.130 (-0.12)	0.178 (0.14)	0.199 (1.41)	0.194 (1.35)	0.197 (1.38)
AggIVol _{t-1}	1.782 (0.75)	5.182** (2.12)	7.910*** (2.86)	11.812*** (3.27)	11.955** (2.62)	11.052** (2.40)
MktVol _{t-1}				-10.551 (-1.63)		-12.937 (-1.18)
TotVol _{t-1}					-5.291 (-1.11)	2.148 (0.27)
MomRet _{t-1}	0.159 (1.18)	0.011 (0.08)	-0.044 (-0.32)	-0.095 (-0.68)	-0.059 (-0.42)	-0.100 (-0.704)
1YMktRet _{t-1}	0.013 (0.32)	-0.008 (-0.17)	-0.029 (-0.58)	-0.086 (-1.43)	-0.050 (-0.95)	-0.090 (-1.44)
DP _{t-1}		12.737 (0.73)	-8.186 (-0.41)	-11.234 (-0.57)	-3.379 (-0.16)	-13.875 (-0.62)
Term _{t-1}		-27.663* (-1.93)	-39.880** (-2.61)	-16.789 (-0.81)	-27.028 (-1.41)	-16.785 (-0.80)
Def _{t-1}		-31.669** (-2.51)	-32.627** (-2.59)	-26.252** (-2.12)	-34.788*** (-2.78)	-23.753 (-1.53)
TB1Y _{t-1}			-11.066* (-1.91)	-6.775 (-1.08)	-7.206 (-1.07)	-7.372 (-1.10)
Adjusted R ²	0.00	0.16	0.21	0.24	0.22	0.23